

Ogwi — Developer Reference

Handover Document 3 of 4 · v2.1 (context edition) · August 2026 · Non-authoritative quick reference. The Functional & System Specification (Document 2) is the sole normative authority, including its Part C (Architecture & Governance), which this document mirrors for convenience. Section references like B4 point into Document 2. **Any discrepancy between this document and Document 2 resolves in favour of Document 2.**

How this document is used, day to day: \$1 when a term is unfamiliar; \$2 pinned to the wall — every new data flow is checked against the map before it is written, per the closure rule; \$3 while implementing the flight log; \$4 whenever anyone asks "can we tune this number?"; \$5 when wiring recomputes; \$6 as the release checklist; \$7 whenever two places seem to disagree about who owns a value; \$8 as the project plan's spine; \$9 when instrumenting. Nothing here is new law — it is the same law, arranged for speed.

1. Glossary

Ogwi — the company and platform. **Oggi** — the inflatable mascot and AI study buddy (only spelling). **Teach Oggi** — the learner-teaches-the-AI activity. **Flow** — the full-screen learning shell inside a Topic; its three faces are the **Feed** (theory scroll), **Canvas** (conversational activities) and **Quiz frame**; **exam mode** is the strictest face. **Dynamic Island** — the single adaptive control/feedback hub inside the Flow. **Knowledge item** — the atomic schedulable unit of knowledge; owns one memory state. **Rendering** — one askable form of an item (base, variants, ladder versions); grades write to the item. **Format ladder** — multiple-choice — cued — typed production. **Key point** — an atomic, human-reviewed marking unit (~8–20 per topic; critical or supporting) used by blurting, Teach Oggi and the Debrief. **Cue question** — a pre-authored question pointing at a key point without containing it; powers blurt hints, Oggi's Teach questions, Teaching Notes and Debrief probes. **Mapping table** — item ↔ objective ↔ module ↔ blueprint weight; the hub every scoring system reads. **FSRS (Free Spaced Repetition Scheduler)** — the published open-source spaced-repetition algorithm the scheduler runs unmodified; the D/S/R model, update equations and default parameters are FSRS's published ones (release version pinned in SchedulerConfig). **R / Stability / Difficulty** — the per-item FSRS memory state; R is current recall probability. **Due** — projected $R \leq 0.90$. **Remediation** — the repair state entered on any incorrect; exited by two correct retrievals on two renderings on two days. **Opener** — the 4–8-question review that starts every session. **"Fixing gaps in: [Module]" / "Keeping fresh: [Module]"** — the two labelled quiz types the engines place in Practice → Recommended. **Mastery** — blueprint-weighted aggregate of item R (live layer) with a kind published display layer. **Pass mark** — the qualification's real pass threshold; the single bar for mastered states, Biggest Opportunity and readiness. **Readiness** — $P(\text{pass})$: the calibrated odds of passing if sat soon. **Forecast** — the expected programme-completion date; also sets the spacing horizon. **Litre / Helium** — the sole currency (1L = 1ft); **LitreConfig** prices everything. **Flight** — a continuous airborne period; **liftoff** at 100L; **touchdown** at 0; **episode** — a continuous period below 100ft (push allowance resets only on climbing back above 100). **The Skyline** — 500ft; **the Ceiling** — 1000ft; **Orbit** — the latched

"??? ft" display state above the cap. **Pump** — a litre-crediting event (every exit pumps). **Publish point** — a moment display values recompute (\$5). **Blueprint** — the awarding body's published exam structure and weights.

2. Canonical entity & data-flow map

The hub: the mapping table (owned by B1, authored by B10).

CONTENT GRAPH (B10) —views→ mapping table · rubrics · cue banks ·
Teaching Notes ·
renderings · lesson chunks ·
confusable edges · templates

RUNTIME STATE	owner
per-item memory states (D,S,R)	B4 (read by B1, B2)
remediation bank + rungs	B3
flight event log	B9
notes store (Oggi)	B11
published mastery/odds/forecast	B1 / B2

FLows

graded answer (any surface) → one review event → B4 state update
AND B3 entry/exit
blurt / Teach / Debrief scoring → per-key-point writes → same
review-event path
B4 due sets + B3 batches → openers & Recommended (composed by B5)
litre events (B8, source-itemised) → B9 pumps → altitude,
records, awards
forecast (B2) → spacing horizon → B4 interval ceilings
engine states + graph → Oggi context frames (B11); Oggi writes
only notes + ordinary grades
published values only → Progress, Wingman, admin surfaces

Closure rule (release-blocking): no data-flow arrow may exist in code that does not exist on this map; adding one requires a map revision first. Nothing reads a raw store except its owning system.

3. The flight event model (summary; full rules B9)

Append-only log; six primary domain event types, plus the **annul** correction type: **pump** (source-itemised, idempotent ID) · **liftoff** (synchronous with its crossing pump) · **touchdown** (materialised once at first discovery; atomic one-per-flight; effective timestamp = solved crossing) · **award-earned** (same pattern, one per award) · **notification-sent** · **config/timezone markers**. Ordinary reads append nothing; displays compute at render from cache + now. Bi-temporal: effective timestamp for physics, append sequence for ties/races; late offline pumps insert at true time with forward recompute; falsified transitions get **compensating annul events** — corrections only ever extend flights. One pure evaluator function owns all decay maths.

4. Configuration registry

Config	Owner	Contents (examples)
LitreConfig	B8	class rates 5/7/1 · unit rates · bonus values · +30L session cap · premiums 50/20/10 · PREMIUM_MIN_ANSWERED 70% · guarantee cap 60
PhysicsConfig	B9	leak 90L/day · 50%/day · threshold 200 · liftoff 100 · cap 1000 · push 75L/18h · quiet hours
SchedulerConfig	B4/B2	desired retention 0.90 · fuzz $\pm 5\%$ · floor 1d · horizon 45d/12mo/3mo default · optimizer trigger RMSE 0.07 · FSRs version pin
MasteryConfig	B1	displayed-decline half-life 7d · daily-rollover timing
OggiConfig	B11	unstick 2-wrong / 90s / lap-repeat · ≤ 2 offers/session · greet 1/session
ReadinessConfig	B2	14d projection · clamp [0.7,1.1] · σ floor · 20% display floor · 80% celebration · band thresholds
Content-graph versions	B10	one immutable version per syllabus edition

Governance law (all configs): immutable, timestamped, versioned documents; forward-only application; **every event stamped with the versions in force**; replay selects the version for its era; physics-affecting changes require cohort-scale comparison or controlled experiment before adoption (versioning gives attribution, never causation); no config is learner-facing.

5. Publish-point table

Moment	Recomputes
Every graded answer	item memory state (B4) · remediation entry/exit (B3)
Every pump / exit / read	flight state at render (B9)

Moment	Recomputes
Session end / Practice-run end	published mastery (B1) · Biggest Opportunity (B1) · odds + certainty + next action (B2) · forecast (B2)
Daily rollover (inactivity)	the same four, once
Blurt/Teach/Debrief scoring moment	their single engine write
Engine batch-queue changes (B3/B4 thresholds or ready-days)	Recommended's contents (A9 §4)
Never mid-activity	any published display value

The publish-point taxonomy itself is law; only the daily rollover's clock time is tunable (MasteryConfig).

6. Cross-system invariants (one always-green suite; any red blocks release)

1. Litres never subtract.
2. Ordinary reads append nothing to the event log.
3. Flight-history corrections only ever extend flights.
4. One serve → both engines written; an item is never double-served.
5. No learner-surfaced prediction is computed from a smoothed input.
6. Mastery can fall only via a failed previously-known item or time decay.
7. Nothing expires, resets, or is a deadline.
8. Every learner-facing artifact passed the human review gate.
9. Every event carries its config-version IDs.
10. Exam mode contains no aid machinery.
11. Sensitive content is structurally unstorable.
12. Attempt counts surface nowhere.

7. Single source of truth

Value / state	Owner (authoritative)	Everyone else
Content, rubrics, cue questions, notes, edges, templates, weights	the versioned content graph (B10)	views only
item ↔ objective ↔ module ↔ weight	the mapping table (B1)	read-only
Per-item memory (D, S, R)	scheduler state (B4)	derived reads
Remediation membership + rung	the bank (B3)	via its 3-signal interface
Published mastery / states / Biggest Opportunity	B1 publishing	display
Odds, certainty, next action, forecast, horizon	B2	display / B4 ceiling
Litre pricing	LitreConfig via B8	none
Flight history, records, awards, episodes	the event log (B9)	caches/views
Oggi's notes	the notes store (B11)	the panel renders it

Value / state	Owner (authoritative)	Everyone else
		verbatim
Voice & copy	the team's tone-of-voice documents	enforcement derives from them
Pass mark	the qualification record (from the awarding body)	all thresholds

8. Implementation checklist (build order)

1. **Content graph + mapping table** with all build-time gates; author one pilot module end-to-end (pages, items, ≥ 3 renderings, key points + cues, notes, edges, template).
2. **Scheduler states + review-event path** (correct=Good / incorrect=Again; defaults; 0.90; fuzz/floor).
3. **Mastery layer:** aggregation, decay, publishing (jump-up / 7-day-half-life down), states vs pass mark, drag; ship the downside-ledger test suite.
4. **Adaptive bank:** entry/exit, ladder, batching, gap-quiz composition, wrong-pool, the 3-signal interface.
5. **Composition:** openers (never-empty, caps), slot-fill (retrieve-before-complete), routing, resume, recall day.
6. **Rubric activities:** blurt scoring pipeline (+voice), Teach (ratchet, chips, meter, static notes, single-write close), Debrief stages.
7. **Economy:** LitreConfig, class pricing at serve, bonuses + cap, assessment payouts + 70% rule, the grounded-liftoff guarantee; run the calibration simulation gates.
8. **Flight:** the pure evaluator; the six-type event log with atomic materialisation, bi-temporal replay, annul corrections; displays at render; the push; records; Orbit; award semantics.
9. **Readiness:** $\mu/\sigma/\Phi$, calibration ratios, bands, unlock checklist, display rules, celebration, forecast + horizon coupling.
10. **Oggi:** frames + logging, contracts + ToV enforcement, ladder, greet, unstick, prompt programs, notes store + panel + sensitive filter, capability boundary.
11. **Surfaces:** the shell, Practice, Progress, onboarding, Wingman, Arrival/referrals, money, awards UI.
12. **The invariant suite + the closure-rule review;** wire analytics (calibration pairs, RMSE, flight metrics, litre parity) — then launch gates: LitreConfig simulation green, invariants green, pilot module through every gate.

9. Analytics quick list

Review events (per grade) · litre events (source-itemised) · flight events (all types) · published-value histories · programme completion rate · experiment assignment/version (launch A/Bs, onboarding pulse) · readiness-calibration pairs (odds-at-sitting vs self-reported outcome) · scheduler RMSE per cohort · unstick/greet offer-accept-dismiss · frame + contract-version logs (QA replay) · QA-queue signals (reports, tripwire, breakdown, exhaustion) · the Wingman instrumentation suite (A15: invite funnel, digest delivery/opens, encouragement counts, link

visits, disclosure-toggle usage, mute/remove/step-down, Tracker opt-ins, quiet-week sends — all content-free). Never collected or displayed: hours, videos completed, login counts, attempt counts, "most flights."